A dual linear programming bound for sphere packing in dimension 36

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Abstract

We construct an explicit dual-feasible point for the Cohn–Elkies linear program in dimension 36, built from the space of weight-18 modular forms for $\Gamma_0(24)$ following the method of Cohn and Triantafillou. The certificate shows that the two-point linear programming bound on the sphere packing density in dimension 36 exceeds the density of the best packing currently known — the Kschischang–Pasupathy packing, of center density $2^{18}/3^{10}$ — by a factor of at least 32.91. In particular, no Cohn–Elkies auxiliary function can certify the best known packing in dimension 36 as optimal. To our knowledge this is the first such dual bound in any dimension above 32, extending the table of Cohn–Triantafillou ($d = 12, 16, 20, 28, 32$), Li ($3 \leq d \leq 13$), and de Courcy-Ireland–Dostert–Viazovska ($d = 6$). The certificate is exact: the dual point is a rational vector, coefficient nonnegativity is verified by exact arithmetic up to $n = 800$, and eventual positivity of the two relevant q-expansions is proved via an explicit Deligne-type tail bound whose constant is certified with outward-rounded interval arithmetic. Two methodological points may be of independent interest: a constraint-generation (cutting-plane) formulation of the exact rational LP, which is what makes an exact vertex whose Eisenstein data supports the tail argument reachable; and a sharpened, lift-aware form of the Deligne bookkeeping constant that discounts deep oldform lifts, without which the finite verification in dimension 36 fails (the crossover moves past the verified window).

1 Introduction

The Cohn–Elkies linear programming bound [1] is the strongest known general upper bound on sphere packing densities in moderate dimensions, and it is famously sharp in dimensions 8 and 24 [2, 3] (and trivially sharp in dimension 1). In every other dimension its sharpness is open, and the question splits into two inequivalent problems. Write $\delta_{\mathrm{LP}}(d)$ for the optimum of the Cohn–Elkies program (normalized as a center density in the sense of [13]) and $\delta_*(d)$ for the best packing density currently known.

- A *dual bound* is a certified lower bound $\delta_{\mathrm{LP}}(d) \geq L$. If $L > \delta_*(d)$, then no Cohn–Elkies function can prove the optimality of the best known packing: either the LP bound is not sharp, or a denser packing than currently known exists. Cohn and Triantafillou [4] constructed such points from modular forms in dimensions $12, 16, 20, 28, 32$; Li [5] gave a different (discrete-reduction) construction covering $3 \leq d \leq 13$; de Courcy-Ireland, Dostert and Viazovska [6] extended the modular-form method to dimension 6.

- *Strict non-sharpness* is the stronger statement $\delta_{\mathrm{LP}}(d) > \delta_{\max}(d)$, where $\delta_{\max}$ is the true optimal density. This requires, in addition to a dual lower bound $L \leq \delta_{\mathrm{LP}}(d)$, an *independent* upper bound $U \geq \delta_{\max}(d)$ with $U < L$. In the published literature this is established only for $d \in \{3, 4, 5, 6, 12, 16\}$: for $d = 12, 16$ by combining the Cohn–Triantafillou dual points with the three-point bounds of Cohn–de Laat–Salmon [7], and for low dimensions by Li's dual bounds against classical upper bounds [5, 6].

This paper proves a statement of the first kind in dimension 36, the first above 32.[1]

Theorem 1.1. $\delta_{\mathrm{LP}}(36) \geq B = 32.91044\ldots \times \dfrac{2^{18}}{3^{10}} > 146.1036$, *where B is the explicit rational number given in* (4)*, with outward decimal enclosures* $B \in [146.1036734821, 146.1036734822]$ *and* $B/(2^{18}/3^{10}) \in [32.91044546, 32.91044547]$. *In particular the Cohn–Elkies linear programming bound in dimension* 36 *cannot certify the Kschischang–Pasupathy packing [8], of center density* $2^{18}/3^{10} = 4.4394\ldots$*, as an optimal sphere packing.*

The witness is an explicit dual-feasible pair of modular forms $(g, \tilde{g})$ of weight 18 for $\Gamma_0(24)$, with $\tilde{g}$ the Fricke transform of g; the required nonnegativity of the two q-expansions is verified exactly for $1 \leq n \leq 800$ and proved for all $n > n_0$ with an explicit crossover $n_0 \leq 63$, closing the gap.

Dimension 36 is a natural target: the weight $k = d/2 = 18$ is even, so no character twists are needed, and 36 is the first dimension beyond the Cohn–Triantafillou table where $\dim S_{d/2}$ obstructions do not apply. It is also qualitatively different from the dimensions ≤ 32: the ratio $\delta_{\mathrm{LP}}/\delta_*$ is much larger (our certificate alone gives 32.9; the best numerical two-point upper bound is $\approx 52 \times \delta_*$), reflecting how weak the record packing is relative to the LP bound in this range. Ironically this makes the *dual* statement stronger and the *strict* question harder; see §5.

Why the construction is not routine. Running the Cohn–Triantafillou method at $d = 36$ meets two obstacles that do not occur (or are mild) at $d \leq 32$.

First, the naïve LP relaxation (maximize the value subject to nonnegativity of finitely many coefficients) produces optima whose q-expansions turn negative just past the enforced range, on the residue classes $n \equiv 3, 9, 15, 21 \pmod{24}$; while the fully constrained problem (nonnegativity through the eventual-positivity crossover, hundreds of rows with coefficients of size $\sim n^{17}$) is numerically singular for floating-point solvers and prohibitively large for exact dense simplex. We resolve this by *constraint generation*: solve the small relaxation exactly in rational arithmetic, evaluate all violated rows exactly, add them, and re-solve. In dimension 36 this converges in a single round (all violations lie in the four classes above and are eliminated together), producing an exact rational vertex — within an explicitly described 29-dimensional search subspace, see §3.1 — whose q-expansions are nonnegative far beyond the enforced range (§3).

Second, the eventual-positivity verification of [4, §5] requires an explicit constant C with $|c_n| \leq C\,\sigma_0(n)\,n^{(k-1)/2}$ for the cuspidal parts of *both* g and $\tilde{g}$. The natural constant — the ℓ^1-norm $\sum |\lambda_{f,e}|$ of the coordinates in the newform-lift basis — turns out to be *lossy* in a way that matters here: our certificate's cuspidal part is dominated (99.7% of the norm) by a single deep

[1] Novelty scope and search date: we searched (2026-07-11, re-checked 2026-07-14) for certified dual lower bounds on δ_{LP} of any kind — Cohn–Elkies-type dual-feasible distributions/modular-form points as in [4, 6], discrete reductions as in [5] — and for LP non-sharpness claims by any method, in dimensions > 32; principal sources: citation graphs of [4] (OpenAlex/Crossref), arXiv full-text sweeps 2019–2026, and Cohn's sphere-packing bounds tables and publication list. See §5 for the two near-misses ([11], [9]).

oldform lift, the $e = 24$ shift of the level-one newform, and with the naïve constant the crossover exceeds the verified window (uniform crossover ≈ 1372; even the divisor-free lower bound is $\approx 827 > 800$, see §4). The lift-aware constant $C_w = \sum_{f,e} |\lambda_{f,e}| \, e^{-(k-1)/2}$ (Lemma 4.1), equally rigorous and elementary, is smaller here by ten orders of magnitude and closes the verification with a margin of 7×10^8 (§4).

A further point, absent from dimensions where the certificate has a sparse structure, is that *both* cusps genuinely matter: not only the Fricke transform $\tilde{g}$ but also g itself has a razor-thin Eisenstein component ($e_1(g) \approx -3.9 \times 10^{-11}$), so the $a_n \geq 0$ side needs the same Deligne-type argument as the $b_n \geq 0$ side, with its own constants (§4).

Rigor. The dual vertex is an exact rational vector (the entries have ≈ 128-digit numerators). All coefficient scans up to $n = 800$ (and, in a second implementation, to $n = 1100$) are exact. The newform decompositions of both cuspidal parts are validated by an exact identity: the reconstruction of the q-expansion from the computed coordinates agrees with the exact coefficients *as rational numbers* (deviation identically 0) up to $n = 800$. The tail constants C_w are certified as upper bounds using outward-rounded interval arithmetic, with the algebraic Hecke eigenvalues enclosed by exact Sturm isolating intervals. Independent implementations (exact characteristic-polynomial factorization vs. high-precision simultaneous diagonalization) produce identical constants. All code, receipts, and the exact certificate data are provided as ancillary files (see the manifest in `ancillary/MANIFEST.md`); a single driver, `ancillary/code/verify_certificate.py`, re-runs every exact-data gate of the verification from the published data — taking the separately certified C_w upper bounds as input; their interval certification is re-run by `d36_cs_certificate.py` — and prints `VERIFIED` only if every gate passes. Numerical-display convention, used throughout: every displayed decimal is either marked *exact*, or is a truncation (rounded toward zero) of an exact rational, or an outward-rounded enclosure/upper bound (so for the constants C_w the displays are certified upper bounds), or — only in the discussion §5 — an explicitly labeled numerical estimate.

2 The Cohn–Elkies program and its modular-form dual

The engine is the following observation of Cohn and Triantafillou [4, Prop. 2.1]: if μ is a tempered distribution on $\mathbb{R}^d$ with $\mu = \delta_0 + \nu$, $\nu \geq 0$, $\mathrm{supp}(\nu) \subseteq \{x \in \mathbb{R}^d : |x| \geq r\}$ for some $r > 0$, and $\hat{\mu} \geq c\,\delta_0$ for some $c > 0$, then the Cohn–Elkies linear programming bound in $\mathbb{R}^d$ is at least $c\,(r/2)^d$. (Any Cohn–Elkies auxiliary function f pairs with μ on both sides of the inequality chain, forcing $f(0)/\hat{f}(0) \geq c\,r^d$ up to normalization.) Sections 2–3 of [4] construct such distributions from modular forms, as follows.

Fix the dimension $d = 36$, weight $k = d/2 = 18$, and level $N = 24$. For a holomorphic modular form $g \in M_k(\Gamma_0(N))$ write $g = \sum_{n \geq 0} a_n q^n$ ($q = e^{2\pi i z}$) and let

$$\tilde{g} \;=\; i^k \left(g\big|_k w_N\right), \qquad w_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}, \qquad (f|_k M)(z) = (\det M)^{k/2}(cz+d)^{-k} f(Mz),$$

be its (normalized) Fricke transform, i.e. $\tilde{g}(z) = i^k N^{-k/2} z^{-k} g(-1/(Nz))$; for our even weight $k = 18$ the factor is $i^{18} = -1$. Then $\tilde{g} \in M_k(\Gamma_0(N))$ as well, and $\tilde{\tilde{g}} = g$. Write $\tilde{g} = \sum_{n \geq 0} b_n q^n$. Given g satisfying the coefficient conditions below, the construction of [4, §§2–3] produces a distribution μ as above with $r = \sqrt{T}$ and $c = (2/\sqrt{N})^{d/2}\, b_0$. Concretely: from any g satisfying

$$a_0 = 1, \qquad a_n = 0 \;(1 \leq n \leq T-1), \qquad a_n \geq 0 \;(n \geq T), \qquad b_n \geq 0 \;(n \geq 0), \tag{1}$$

a dual-feasible point for the Cohn–Elkies program with value (as a center density)

$$B(g) \;=\; b_0 \cdot \left(\tfrac{2}{\sqrt{N}}\right)^{d/2} \cdot \left(\tfrac{\sqrt{T}}{2}\right)^{d}, \tag{2}$$

so that $\delta_{\mathrm{LP}}(d) \geq B(g)$. Here T is a free integer parameter (the cutoff index: the $T-1$ coefficients $a_1, \ldots, a_{T-1}$ are forced to vanish). We take $T = 10$. As a consistency check of the normalization, our implementation of (1)–(2) reproduces the published $d = 12$ values of [4] at both levels: $0.059782 = 1.6141\times$ Coxeter–Todd at $N = 24$ ([4, Table 6.1]) and $0.0624462 = 1.68605\times$ at $N = 96$ ([4, Thm. 1.1]), to the displayed precision.

3 The certificate

3.1 The space, the search subspace, and the reduced LP

$\dim M_{18}(\Gamma_0(24)) = 72$, with an 8-dimensional Eisenstein subspace spanned by $E_{18}(\delta z)$, $\delta \mid 24$, where

$$E_{18}(z) = 1 + \kappa_{18} \sum_{n\geq 1} \sigma_{17}(n) q^n, \qquad \kappa_{18} = -\frac{2k}{B_{18}}\Big|_{k=18} = -\frac{28728}{43867},$$

and a 64-dimensional cuspidal subspace. Candidate forms are the eight Eisenstein series together with the holomorphic eta-quotients $\prod_{\delta|24} \eta(\delta z)^{r_\delta}$ of weight 18 and level dividing 24 (Ligozat's criteria) with exponent bound $|r_\delta| \leq 8$: there are 3044 such eta-quotients, and the 3052 candidates span all of $M_{18}(\Gamma_0(24))$ — an exact certificate: the candidate coefficient matrix truncated at q^{100} has rank 72 modulo the prime $10^9 + 7$, hence exact rank ≥ 72, and $= 72$ by the dimension formula (`d36_span_rank_certify.py` and its receipt in the ancillary files; at $|r_\delta| \leq 6$ only 44 eta-quotients exist, which under-span). All q-expansions used in the construction and verification are computed in exact integer/rational arithmetic.

The linear program is *not* solved over the full 72-dimensional space but over an explicit 29-dimensional subspace V, chosen as follows. The LP window below constrains the coefficients $a_0, \ldots, a_{28}$; column-pivoted QR factorization, applied to the *transpose* of the floating-point candidate coefficient matrix truncated at q^{28} (i.e. to the 29×3052 matrix whose columns are the candidates' truncated coefficient vectors, so that the pivoting selects candidates), selects 29 pivot forms — the truncation has only 29 coefficients $a_0, \ldots, a_{28}$, so at most 29 candidates are linearly independent in this window. The selected forms are 7 Eisenstein series ($E_{18}(\delta z)$, $\delta \in \{1, 2, 3, 4, 6, 8, 24\}$) and 22 eta-quotients; the full list, in pivot order (which fixes the coordinate system used everywhere, including the published certificate), is in the ancillary file `ancillary/certificate_exact_data.txt`, and V is the span of these 29 forms. This pivot selection is the one floating-point step in the construction: it only decides *which* forms to use, and everything downstream treats the selected forms exactly. Restricting the search to $V \subsetneq M_{18}(\Gamma_0(24))$ can only weaken the bound we are able to find, never its validity: Theorem 1.1 requires exhibiting one dual-feasible form, and feasibility of the certificate is verified unconditionally (§4).

Within V, the 9 homogeneous conditions $a_1 = \cdots = a_9 = 0$ cut out a nullspace of exact dimension $r = 29 - 9 = 20$: in exact rational arithmetic, the 9×29 coefficient matrix of $(a_1, \ldots, a_9)$ has rank 9, and the 10×29 matrix of $(a_0, a_1, \ldots, a_9)$ has rank 10 — so a_0 is not identically zero on the nullspace and the normalized family is nonempty (both ranks are recertified by the verification driver). The LP variables are the $r = 20$ nullspace coordinates, with the normalization $a_0 = 1$ imposed as an equality constraint of the program (the feasible forms thus lie in a 19-dimensional affine slice of the nullspace); in these variables we solve

$$\max\ b_0 \quad \text{s.t.} \quad a_n \geq 0\ (10 \leq n \leq 28), \qquad b_n \geq 0\ (n \in W), \tag{3}$$

with a working set W of Fricke-side rows, in exact rational arithmetic. All optimality statements in this section are relative to V: we make no claim about the optimum of (3) over the full space $M_{18}(\Gamma_0(24))$, and none is needed for Theorem 1.1.

3.2 Constraint generation

With $W = \{1, \dots, 28\}$ the exact optimum of (3) (over V) is $36.3470 \times 2^{18}/3^{10}$, but its Fricke expansion has 45 negative coefficients in $1 \leq n \leq 300$, all in the classes $n \equiv 3, 9, 15, 21 \pmod{24}$, the first at $n = 33$: the unconstrained optimum overshoots into infeasibility, and its Eisenstein data forces $b_n < 0$ for infinitely many n, so no finite verification can rescue it: for that vertex the binding class-3 Eisenstein combination $r_3 = e_1 + e_3/(1 + 3^{17})$ is *positive* ($\approx +4.3 \times 10^{-13}$; exact value in the construction receipt), so for $n = 3p$, $p > 3$ prime, the Eisenstein term equals $\kappa_{18}\, r_3\, (1 + 3^{17})\, \sigma_{17}(p) < 0$ (recall $\kappa_{18} < 0$), of size $\asymp n^{17}$, while the cuspidal part is $O(\sigma_0(n)\, n^{17/2})$ by Deligne's bound — so $b_n \to -\infty$ along that class. Adding all 45 violated rows to W ($|W| = 73$) and re-solving exactly gives a vertex, optimal for the enlarged program over V, with *no* violations: re-scanning exactly, $a_n \geq 0$ and $b_n \geq 0$ for all $1 \leq n \leq 800$ (zero negative coefficients on either side, including the un-enforced range $300 < n \leq 800$). The cutting-plane process converges in one round (a statement about this initialization and violation scan, not a general convergence claim). Exactly,

$$B \;=\; b_0 \cdot \frac{5^{18}}{2^{27}\, 3^9}, \qquad b_0 = \frac{p}{q} \ \ \text{(exact rational, in lowest terms)}, \tag{4}$$

where $\frac{5^{18}}{2^{27}3^9} = \frac{3814697265625}{2641807540224}$ is the exact value of $(2/\sqrt{24})^{18}(\sqrt{10}/2)^{36}$ and

$$\begin{aligned} p = {} & 95066\,91735\,60589\,08133\,22373\,57126\,88188\,23298\,71412\,02030\,77849\,06752\,56657 \\ & 93709\,84887\,76440\,11565\,11331\,34854\,11947\,23969\,08932\,10895\,90633\,472, \\ q = {} & 93956\,57537\,80418\,71093\,01784\,65357\,83534\,26840\,65696\,00550\,96336\,00075\,25366 \\ & 95749\,03464\,45951\,04757\,46960\,28937\,05377\,61585\,96625\,88923\,78832\,3, \end{aligned}$$

(123 and 121 digits; the fraction for B itself, and outward decimal enclosures $b_0 \in [101.1817608012, 101.1817608013]$, $B \in [146.1036734821, 146.1036734822]$, are in the ancillary file). Exact rational arithmetic in the solve is essential: at this conditioning, floating-point and even 300-digit fixed-precision simplex return phantom vertices (violating $a_0 = 1$) once rows with coefficients of size 10^{30} enter the basis.

Remark 3.1. The certificate data — the explicit list of the 29 basis forms, the 20 exact rational coordinates of the vertex in the (pivot-ordered) reduced basis, the coordinate-to-form map, the exact rational b_0 and B, the exact Eisenstein coefficients, and all constants below — are in the ancillary file `ancillary/certificate_exact_data.txt`; the construction and verification receipts are `ancillary/receipt_d36_cutgen.txt` and `ancillary/receipt_d36_cs.txt`, and the single-driver verification is `ancillary/code/verify_certificate.py` (§5).

4 Eventual positivity

It remains to prove $a_n \geq 0$ and $b_n \geq 0$ for all $n > 800$. Decompose both forms into Eisenstein and cuspidal parts; we treat $\tilde{g}$ (the binding side) and note the same argument for g with its own constants.

4.1 Eisenstein main term, per residue class

Write the Eisenstein component of $\tilde{g}$ as $\sum_{\delta|24} e_\delta\, E_{18}(\delta z)$. The exact rational e_δ (computed by an exact projector, validated as explained below) have the numerical values

$$\begin{aligned} e_1 &= -4.197922 \cdot 10^{-13}, & e_2 &= -2.023980 \cdot 10^{-14},\\ e_3 &= +5.361610 \cdot 10^{-5}, & e_4 &= +5.776815 \cdot 10^{-3},\\ e_6 &= -18.24675, & e_8 &= -27.55171,\\ e_{12} &= +43.87748, & e_{24} &= +103.0969. \end{aligned}$$

For $n \geq 1$ the Eisenstein coefficient is $\kappa_{18} \sum_{\delta|\gamma} e_\delta\, \sigma_{17}(n/\delta)$, where $\gamma = \gcd(n, 24)$. Using $m^{17} \leq \sigma_{17}(m) < \zeta(17)\, m^{17}$ together with the rational bound $\zeta(17) < 1 + 2^{-17} + 2^{-16}/16$ (compare the sum with $\int_2^\infty x^{-17}dx$), and bounding each term from below according to its sign (note $\kappa_{18} < 0$, so $\kappa_{18}e_\delta > 0$ exactly when $e_\delta < 0$), we get for every $n \geq 1$ in the class γ:

$$\text{Eisenstein}(n) \;\geq\; c_E(\gamma)\, n^{17}, \qquad c_E(\gamma) \;=\; \sum_{\substack{\delta|\gamma \\ e_\delta<0}} \frac{\kappa_{18}e_\delta}{\delta^{17}} \;+\; \left(1 + 2^{-17} + \tfrac{2^{-16}}{16}\right) \sum_{\substack{\delta|\gamma \\ e_\delta>0}} \frac{\kappa_{18}e_\delta}{\delta^{17}}, \tag{5}$$

an exact rational number for each γ. Evaluating exactly: all sixteen constants (eight classes, both sides) are *positive*, with both minima on the class $\gamma = 3$:

γ	1	2	3	4	6	8	12	24
$c_E(\gamma)$, $\tilde{g}$-side ($\times 10^{-13}$)	2.749	2.749	**0.0301**	0.547	7.089	0.627	4.887	4.967
$c_E^{(a)}(\gamma)$, g-side ($\times 10^{-11}$)	2.555	4.731	**0.506**	2.921	2.682	2.921	0.873	0.873

$$c_E \;:=\; \min_{\gamma|24} c_E(\gamma) \;=\; c_E(3) \;=\; 3.0197202 \cdot 10^{-15} \quad (\tilde{g}\text{-side}), \tag{6}$$

and analogously $c_E^{(a)} = c_E^{(a)}(3) = 5.0672217 \cdot 10^{-12}$ for g (again the minimum over all eight classes). All sixteen constants are the $\zeta(17)$-guarded floors of (5) — hence slightly below the raw leading Eisenstein coefficient $\kappa_{18} \sum_{\delta|\gamma} e_\delta/\delta^{17}$ of each class (the quantity tabulated in the verification receipt) — and are exact rationals; the table and (6) display truncations, and the exact numerators/denominators are printed in the ancillary file `ancillary/guarded_cE.txt` (output of `code/d36_guarded_cE.py`). The thinness of these constants — forced by the active constraints at the optimal vertex of (3) — is what makes the tail argument in dimension 36 delicate: the certificate value $b_0 \approx 101.2$ exceeds the per-class Eisenstein floor constants c_E by factors of $\approx 2 \times 10^{13}$ (g-side) and $\approx 3 \times 10^{16}$ ($\tilde{g}$-side).

4.2 The cuspidal remainder and a lift-aware Deligne constant

Let $S = \tilde{g} - (\text{Eisenstein part}) = \sum c_n q^n \in S_{18}(\Gamma_0(24))$, a space of dimension 64. The newform/oldform basis consists of the lifts $f(ez)$, where f runs over the normalized newforms of level $M \mid 24$ and $e \mid 24/M$; the newspace dimensions are

$$(\dim S_{18}^{\mathrm{new}}(\Gamma_0(M)))_{M|24} = (1,1,3,2,3,4,2,9) \quad \text{for } M = (1,2,3,4,6,8,12,24),$$

with lift multiplicities $(8,6,4,4,3,2,2,1)$, summing to 64. We compute the exact coordinates $\lambda_{f,e}$ of S in this basis (Galois orbits handled by exact characteristic-polynomial factorization of the Hecke action; bad-prime lifts separated by U_2, U_3 and leading q-exponents) and *validate the decomposition exactly*: the reconstruction $\sum \lambda_{f,e} f(ez)$ agrees with S coefficient-by-coefficient, as exact rationals, for all $n \leq 800$ (deviation identically zero; the Sturm bound for $M_{18}(\Gamma_0(24))$ is 72, so agreement to 800 is a highly overdetermined identity). This validation is an identity between *rational* objects — each Galois-orbit contribution $\sum_{f \in \text{orbit}} \lambda_{f,e} f(ez)$ has rational coefficients and is computed by exact rational linear algebra — so it is independent of the interval layer described below, which enters only through the absolute values $|\lambda_{f,e}|$ of the individual conjugates inside a multi-dimensional orbit. As a further check, the level-one component is the unique normalized cusp form ΔE_6 of weight 18, and the computed eigenvalues match $a_2 = -528$, $a_3 = -4284$, $a_5 = -1025850$, $a_7 = 3225992$ exactly.

Lemma 4.1 (lift-aware Deligne constant)**.** *Let $S = \sum_{f,e} \lambda_{f,e} f(ez) \in S_k(\Gamma_0(N))$ with f normalized newforms (each Galois orbit expanded into its individual complex embeddings, so that every $|\lambda_{f,e}|$ is the modulus of a single complex coordinate). Then for all $n \geq 1$,*

$$|c_n| \;\leq\; C_w\, \sigma_0(n)\, n^{(k-1)/2}, \qquad C_w \;=\; \sum_{f,e} \frac{|\lambda_{f,e}|}{e^{(k-1)/2}}.$$

Proof. By the divisor bound $|a_f(m)| \leq \sigma_0(m) m^{(k-1)/2}$, valid for every $m \geq 1$ and each normalized newform — Deligne's estimate with the Hecke recursion at the unramified primes $p \nmid M$, and the standard newform bound $|a_f(p)| \leq p^{(k-2)/2} \leq \sigma_0(p)\, p^{(k-1)/2}$ at the ramified primes $p \mid M$. The coefficient of q^n in $f(ez)$ is $a_f(n/e)$ if $e \mid n$ and 0 otherwise, and for $e \mid n$, $\sigma_0(n/e) \leq \sigma_0(n)$ and $(n/e)^{(k-1)/2} = n^{(k-1)/2} e^{-(k-1)/2}$. Summing over the pairs (f, e) with $e \mid n$ gives the claim. $\square$

The point of the lemma is quantitative. The naïve constant $\sum_{f,e} |\lambda_{f,e}|$ equals 1.894×10^{10} for our S: it is dominated (99.7%) by the single coordinate $\lambda_{\Delta E_6, 24} \approx -1.89 \times 10^{10}$ on the deepest oldform lift, and with it the verification over $n \leq 800$ fails: under the uniform inequality actually used below ($\sigma_0(n) \leq 2\sqrt{n}$, crossover $(2C/c_E)^{1/8}$) the naïve crossover is $n_0 \approx 1372$; even discarding the divisor factor altogether — which gives a valid *lower* bound on the crossover, since $\sigma_0(n) \geq 1$ — positivity cannot be established below $(C/c_E)^{2/17} \approx 827 > 800$. The lift-aware constant suppresses the deep-lift term by $24^{17/2} \approx 5.4 \times 10^{11}$:

$$C_w \;\leq\; 0.358807778339 \quad (\tilde{g}), \qquad C_w^{(a)} \;\leq\; 0.306329333530 \quad (g), \tag{7}$$

certified as upper bounds by outward-rounded interval arithmetic, as follows.

Exact part. The coordinates on rational (one-dimensional Galois orbit) blocks, including the dominant level-one tower, are exact rational numbers (31 of the 64 lifts); their contribution to C_w is exact up to the outward-rounded enclosure of $e^{-17/2}$.

Interval part (the 33 algebraic coordinates). The coordinates themselves are exact algebraic numbers, specified by exact polynomial and embedding data; certified real intervals are used only to bound their absolute values for C_w. The residual $R = S -$ (rational-block part) is itself an exact rational q-expansion lying in the 33-dimensional span of the multi-dimensional orbits. Each Hecke eigenvalue of such an orbit is enclosed by an exact isolating interval with rational endpoints, produced by a Sturm-sequence root-isolation computation for the exact characteristic polynomial (isolation width $\leq 10^{-40}$; this fixes each real embedding unambiguously — not to be confused with the Sturm bound for modular forms used above), the 33 lift q-vectors are propagated as outward-rounded interval vectors, and the coordinates are obtained from a sound interval Gaussian elimination (`mpmath.iv` throughout, `mpmath` version pinned in the ancillary `requirements.txt`) of the 33×33 system on 33 determining rows drawn from $n \in \{117, \ldots, 170\}$ (the exact row list is printed in the verification receipt); the rows are chosen by a floating-point pivoting pass, which affects only *which* rows are used, not the soundness of the interval solve. A pivot interval straddling zero would make the enclosures blow up and the certificate comparison fail; the realized enclosure widths $\sim 10^{-50}$ therefore also certify the nonsingularity of the chosen system, and the verification receipt records the minimal pivot magnitude (bounded away from zero) together with dyadic-rational outward enclosures for both constants (7). Cross-checks: the bound is stable to 15 digits across four disjoint determining windows; artificially degrading the working precision widens the enclosure without ever breaking the bound (confirming outward rounding); the interval coordinates enclose the coordinates computed by the independent second implementation (§5), which reproduces (7) to all displayed digits; and the 33×33 interval solve together with the $|\lambda_{f,e}|$-aggregation is re-verified independently in Arb ball arithmetic (`python-flint`), consuming the exported exact dyadic endpoints (`d36_arb_check.py`; Arb's certified ball solve fails on a singular ball matrix, so its success certifies nonsingularity independently of the pivot log, and its exact dyadic upper bound on the algebraic-remainder sum agrees with the `mpmath.iv` value to all displayed digits — the rational-block part of (7) is exact and common to both computations).

4.3 Closing the certificate

Combining (6), Lemma 4.1, (7) and $\sigma_0(n) \leq 2\sqrt{n}$: for n on any residue class,

$$b_n \;\geq\; c_E\, n^{17} - 2\, C_w\, n^9 \;>\; 0 \qquad \text{whenever } n^8 > \frac{2C_w}{c_E},$$

i.e. for all $n \geq n_0$, where $n_0 = 63$ is the smallest integer with $c_E\, n^8 > 2C_w$ — certified by an exact rational comparison, with C_w replaced by its certified upper bound (7); on the g-side, $n_0^{(a)} = 25$. Since $a_n \geq 0$ and $b_n \geq 0$ have been verified exactly for $1 \leq n \leq 800$, both hold for all n, and $(g, \tilde{g})$ satisfies (1). Theorem 1.1 follows from (2) and (4). □

The margins are enormous ($2C_w/c_E$ would have to exceed its actual value by a factor 7×10^8 before n_0 reached 800); the certificate is in no sense borderline once the correct constant is used. The role of the verification window $n \leq 800$ (rather than, say, $n \leq 63$): the window was fixed before the tail constants were computed, and with the naïve unweighted constant it is insufficient — the uniform crossover is $n_0 \approx 1372$, and even the divisor-free lower bound on the crossover, $(C/c_E)^{2/17} \approx 827$, already exceeds 800 — which is what motivated Lemma 4.1; retaining the full window makes the exact reconstruction identity of §4 overdetermined by an order of magnitude beyond the Sturm bound 72, and a second implementation independently extends the exact sign scans to $n \leq 1100$.

5 Discussion

Strict non-sharpness in dimension 36 is far out of reach. Our dual point shows $\delta_{\mathrm{LP}}(36) \geq 32.91\,\delta_*$ (recall δ_* denotes the best *known* packing density, not the true optimum $\delta_{\max}$), and the best available *numerical* two-point upper bound is $\approx 52.2\,\delta_*$ [1, 4] — a numerical estimate: outside $d \in \{1, 8, 24\}$ the exact optimum of the Cohn–Elkies program is not known, and computed values are approximations from above [4]. Using our present dual point, strict non-sharpness would need an independent upper bound below $32.91\,\delta_*$, i.e. an improvement of $\approx 37\%$ over the two-point bound. For comparison, the only general-purpose technology quantified to improve on the two-point bound in this range — the three-point bounds of [7] — gains 4.85% at $d = 12$ and 0.79% at $d = 16$, and no three-point computation has been reported in any dimension beyond 16 (cf. the computational range of [7]; survey [12]). The paradox of dimension 36 is that the *weakness* of the known record makes the dual statement strong and the strict statement inaccessible: both our dual value and the two-point bound are large multiples of the record precisely because the record is weak; the residual gap between them reflects the suboptimality of our particular certificate (a finite search subspace and the choice $T = 10$), not a property of the problem.

Dimension 40. The same pipeline reaches $d = 40$ (weight 20) in principle, with two new costs: the weight-20 eta-quotients of level 24 span only a 72-dimensional subspace of the 80-dimensional $M_{20}(\Gamma_0(24))$, so the basis must be augmented (e.g. by $E_4 \cdot$ (weight-16 forms)), and the exact arithmetic is heavier (σ_{19} growth). We have not completed a $d = 40$ certificate. We note that Remark 4.3 of [11] asserts that for $d = 32$ and $d = 40$ "LP non-sharpness is established by other means", attributing both to Cohn–Triantafillou; but [4] treats only $d \in \{12, 16, 20, 28, 32\}$ and does not address $d = 40$ (and [11]'s own Table 1 marks the status at $d = 40$ as "expected"). As of July 2026 we have been unable to locate any published dual bound or non-sharpness proof in dimension 40. The modular-bootstrap analysis of [9] (building on the sphere packing–quantum gravity correspondence of [10]) rules out *sharpness* of the LP bound for a range of dimensions by an implied-kissing-number criterion, but reports its first crossings of that criterion only near $d \approx 180$ and, in its authors' own words, "does not give a quantitative improvement"; it therefore neither contains nor implies a dual bound at $d = 36$ or 40.

Methodological remarks. Two ingredients may be reusable. (i) Constraint generation appears to be the right formulation of the exact dual LP whenever eventual positivity must be enforced through a crossover: it keeps every exact solve small (here: one re-solve with 73 rows) while the dense exact LP is intractable and floating-point LP is unsound at these conditionings. (ii) Lemma 4.1 is elementary, but the failure of the unweighted constant in dimension 36 suggests it should be the default bookkeeping in eventual-positivity verifications: deep oldform lifts are exactly the coordinates the unweighted ℓ^1 norm overweights, by the factor $e^{(k-1)/2}$ that the lemma restores.

Reproducibility. The exact verification — the rigorous part of the certificate — uses only exact rational linear algebra, integer q-expansions of eta-quotients, and interval arithmetic (Python: `sympy`, `mpmath`, `fractions`); no computer-algebra system with modular-forms machinery is required. Floating-point routines (`numpy`, `scipy`) are used only in the *construction* step, as heuristics to select the pivot basis (§3.1) and to locate the candidate vertex; they play no role in the exact rigor pass, and every theorem-critical constant reported here is either recomputed exactly or bounded by a certified outward enclosure (the labeled numerical estimates of this section are the only exceptions). The full pipeline, receipts, and the exact certificate are provided as ancillary files, together with an independent second implementation of the newform decomposition (high-precision simultaneous diagonalization) that reproduces the constants (7) to all reported digits; a third, optional CAS-based cross-check — a PARI/GP reproduction (`d36_C_pari_indep.py`, via the native modular-symbols engine, no shared code path) that rebuilds the 64-form newform/oldform lift basis from scratch and again reproduces both constants (7) to all reported digits, along with the newspace dimension table and the level-one eigenvalues; and an independent Arb (ball-arithmetic) re-verification of the theorem-critical interval solve (`d36_arb_check.py`, §4). The ancillary manifest separates the files that constitute the *proof* of Theorem 1.1 from exploratory/receipt material, lists SHA-256 checksums of the data files, and the driver `verify_certificate.py` re-runs the exact-data chain — reduced-basis reconstruction from the published form list, vertex feasibility scans, Eisenstein projection, guarded c_E, tail crossover — from the published exact data alone (with the C_w upper bounds as separately certified inputs), printing `VERIFIED` only if every gate passes.

Data and code availability. The exact certificate data, the construction and verification receipts, and the complete Python pipeline (including the independent second implementation) are provided as ancillary files accompanying this submission; see `ancillary/MANIFEST.md` for the file map and reproduction order.

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References

[1] H. Cohn and N. Elkies, *New upper bounds on sphere packings I*, Ann. of Math. **157** (2003), 689–714. arXiv:math/0110009.

[2] M. Viazovska, *The sphere packing problem in dimension 8*, Ann. of Math. **185** (2017), 991–1015.

[3] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko and M. Viazovska, *The sphere packing problem in dimension 24*, Ann. of Math. **185** (2017), 1017–1033.

[4] H. Cohn and N. Triantafillou, *Dual linear programming bounds for sphere packing via modular forms*, Math. Comp. **91** (2022), 491–508. arXiv:1909.04772.

[5] R. Li, *Dual linear programming bounds for sphere packing via discrete reductions*, Adv. Math. **460** (2024), 110043. arXiv:2206.09876.

[6] M. de Courcy-Ireland, M. Dostert and M. Viazovska, *Six-dimensional sphere packing and linear programming*, Math. Comp. **93** (2024), 1993–2029. arXiv:2211.09044.

[7] H. Cohn, D. de Laat and A. Salmon, *Three-point bounds for sphere packing.* arXiv:2206.15373.

[8] F. R. Kschischang and S. Pasupathy, *Some ternary and quaternary codes and associated sphere packings*, IEEE Trans. Inform. Theory **38** (1992), 227–246.

[9] N. Afkhami-Jeddi, H. Cohn, T. Hartman, D. de Laat and A. Tajdini, *High-dimensional sphere packing and the modular bootstrap*, J. High Energy Phys. **12** (2020), 066. arXiv:2006.02560.

[10] T. Hartman, D. Mazáč and L. Rastelli, *Sphere packing and quantum gravity*, J. High Energy Phys. **12** (2019), 048. arXiv:1905.01319.

[11] J. Zhou, *Cusp form dimensions, lattice uniqueness, and LP sharpness for sphere packing in dimensions 8 and 24*. arXiv:2604.10914.

[12] F. Vallentin, *Conic optimization for extremal geometry*. arXiv:2510.06960.

[13] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999.